

DSA I Lab Assignment

Stack Implementation Using Two Circular Queues

Marks : 15 (Code-10 , Assignment based Quiz - 5)

Language: C++ only

Note: Built-in STL stack, queue, vector, deque, list, or any ready-made container must not be used. Students must implement the required Queue and StackWithQueue classes manually.

Problem:

Write a **single C++ program** that completes the following tasks using user-defined data structures only.

1. Create a **Queue class**. The queue must be implemented as a circular queue using a fixed-size array.
2. The Queue class must contain the functions enqueue() and dequeue(). Other helper functions may be added if needed.
3. Using **two Queue objects**, create a **StackWithQueue** class.
4. The **StackWithQueue class** must contain the functions push() and pop(). Other helper functions such as top(), isEmpty(), or display() may be added if needed.
5. In main(), create a StackWithQueue object and use it where stack operations are needed.
6. Take a decimal number from the user and convert it into binary using stack operations. For this part, complete the function **convertToBinary(int n)**.
7. Take an integer array as input and sort it using stack operations. For this part, complete the function **sortUsingStack(int arr[], int n)**.
8. Print the binary result and the sorted array as shown in the sample output.

Input Format

The program should take input in the following order:

- A decimal integer n.
- The size of the array.
- The elements of the array.

Output Format

- Print the binary representation of the decimal number.
- Print the sorted array in ascending order.

Sample Input

```
10
5
4 1 5 2 3
```

Sample Output

Binary: 1010

Sorted array: 1 2 3 4 5

Submission Instructions

- Submit only one .cpp file.
- Do not use STL stack, queue, vector, deque, list, or similar built-in containers.
- Do not remove the required function names from the starter code.
- Marks will be deducted if the circular queue is not implemented properly.
- Marks will be deducted if stack operations are not implemented using two Queue objects.
- Do not use AI – do it yourself with understanding